

SC475 – Time Series Analysis

Course Placement: Time Series Analysis is an elective course offered to third and fourth-year students of the B.Tech program.

Course format: 3-hour lecture every week

Prerequisites: A basic course in statistics and probability, Linear Algebra, Calculus, and either Nonlinear Dynamics or Modelling and Simulation. Python, R or Matlab programming skills.

Course content: This course introduces students to the essential tools and concepts in time series analysis, covering both time and frequency domains. Time series typically arise in many processes that require consecutive measurements of quantities. Understanding the basic structure and information of a time series requires concepts often used in signal and image processing, statistics, dynamical systems, and statistical physics. The course aims to develop a working foundation at the undergraduate level that enables systematic analysis of time series arising in various disciplines. The course primarily focuses on traditional statistical methods used in time series analysis. In addition, it aims to expose students to modern challenges and selected nonlinear methods that are used extensively in the research literature. We will primarily focus on EEG and financial time series to develop a more hands-on understanding of the concepts discussed in class.

A single textbook is insufficient for this course. Some of the relevant resources that will be referred to are:

Reference Books:

- George E. P. Box, Gwylim M. Jenkins, Gregory C. Reinsel, Greta M. Ljung, *Time Series Analysis. Forecasting and Control*, Fifth Edition, Wiley, 2016
- Donald B. Percival, Andrew T. Walden, *Wavelet Methods for Time Series Analysis*, Cambridge University Press, 2000
- Holger Kantz, Thomas Schreiber, *Nonlinear Time Series Analysis*, Springer, 1999
- R. H. Shumway and D. S. Stoffer, *Time Series Analysis and Its Applications: With R Examples*, fourth edition. Springer 2017.
- C. Chatfield, and H. Xing, *The Analysis of Time Series. An Introduction with R*, seventh edition. Chapman & Hall/CRC 2019.
- *Analyzing Neural time series data*, Theory and Practice, Mike X Cohen, MIT press, 2014.
- *Analysis of Financial time series*, R. S. Shay, Wiley, 2010.
- *Econophysics: An introduction*, Sitabhra Sinha, Arnab Chatterjee, Anirban Chakraborti, and Bikas K. Chakraborti, Wiley 2010.
- *An introduction to Econophysics, Correlations and Complexity in Finance*, R. N. Mantegna and H. E. Stanley, Cambridge, 2004.

Course Outcomes:

After completing the course, the students would be able to analyze time series using variety of methods. In particular, they would be able to do a systematic analysis of time series, especially the ones that fall in the broad domain of complex systems.

- Apply the knowledge of mathematics and computer science to model complex problems in science (**P1,P2,P3,P12**).
- Carry out model based investigations to derive insight into the problem at hand (**P4**).
- Communicate effectively through vivas, course presentation and report writing (**P10**).
- Project Management/Team Work (**P11**).
- Modern Tool Usage (Python/R/Matalb) (**P5**)

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X	X					X	X	X

Course Policy (Evaluation Policy is tentative and will finalized in the first week of the semester)

- **Exam (60%):** Mid-semester (30) + Final (30)
- **Project (40%):** The Project will allow applying the learning in the course to a real data set. A relevant published paper will have to be identified in the first month of the semester. Students will reproduce the key results in the paper and then further analyze the data set. Group projects are encouraged.
- **Attendance Policy:** A minimum of 75% attendance (of all lectures in which attendance is taken) is mandatory. There will be a grade point penalty for students who do not meet the minimum attendance requirement.
- **AI Policy:** Use of AI is permitted for the course project, but all submitted work must be genuinely produced and owned by the students. Each group must be able to explain every detail of their project independently and without relying on AI during evaluation.

Lecture Schedule *

Sl. No.	Description	No. of Lectures
1	Overview and Introductions	2-3
	Examples of time series, Trends in Time Series: Parametric, Non-parametric trends, Periodicity, seasonality, etc	
2	Fundamental Concept from Probability and Random Variables	2-3
	Time Series and Stochastic Processes, Covariance and autocorrelation, etc.	
3	Autocorrelation processes and the spectrum of stationary processes	3-4
	Basic properties and linear processes, Introduction to ARMA models	

4	Linear Stationary Models	4-5
	Autoregressive and moving average processes, as well as mixed autoregressive and moving average processes.	
5	Linear Nonstationary Models	5-8
	Autoregressive Integrated Moving Average processes, Integrated Moving Averages processes	
6	Spectral Analysis of Time Series	5-8
	Spectral Representation and Spectral Distribution: Estimating the Spectrum.	
7	Other Topics	
	Time Series models of Heteroscedasticity, ARCH, GARCH and Stochastic Volatility Models, Nonlinear models, Some concepts from nonlinear dynamical systems, etc.	

* The number of topics covered will depend on the overall enthusiasm and commitment of the class